

Orchestrating Cyber Threat Intelligence using T-Pot and MITRE ATT&CK: From Cyber Attack Data Collection to Advanced Insights

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Abstract—Organizations are facing rising cyberattacks, often without prior defenses, leaving critical infrastructure exposed. This study presents a structured proactive approach to identify possible cyber threats by investigating cyber threat intelligence deploying high-interaction honeypots to attract and analyze nature of the attacks. In this proposed study, we have tailored the deployment for IoT and SCADA systems, integrated with Security Onion for threat intelligence and MITRE ATT&CK framework for measured solutions. Unlike prior static honeypot setups, this cloud-based T-Pot environment enables real-time detection and profiling of attacker tactics, supporting dynamic threat modelling. The configuration reveals high-severity and high-volume threat patterns through live attack surface emulation. Combining honeypots, behavioral analysis, and structured threat mapping advances CTI methodologies and provides a practical framework for resilient infrastructure protection.

Keywords— *Cyber Threat Intelligence, Cyberattack, Cyber security, Honeypot, Internet security, IoT, SCADA systems, Threat analysis*

I. INTRODUCTION

The surge in cyber threats requires organizations to adopt diverse protection strategies [1], yet many lack proactive defenses, exposing them [2]. The 5G and IoT expansion has increased vulnerabilities through widespread device interconnectivity [3]. Industrial Control Systems (ICS), essential to OT in IoT, are common targets due to structural weaknesses [4]. These internet-connected systems, which manage assets like robots and ships [4], have recently seen attacks on programmable logic controllers [4]. Leveraging Cyber Threat Intelligence (CTI) through data analysis helps detect attack patterns early and strengthens defenses [1].

Studies [5], [6] show that the MITRE ATT&CK framework structures adversarial behavior in ICS and IoT systems. As a publicly funded U.S. government resource, it outlines real-world attacker tactics and techniques, aiding threat modelling across sectors [7]. Compared to other frameworks, MITRE ATT&CK offers a more detailed and comprehensive taxonomy, enhancing organizations' ability to understand, detect, and respond to cyber threats. In [8], attacker behavior was examined using the ELK stack on logs from two AWS-hosted honeypots, demonstrating the

usefulness of such data for threat hunting and mitigation. In [9], Conpot's default setup proved insufficient, but minor configuration changes validated through Shodan scans increased ICS traffic, while the default mostly drew FTP connections. In [4], three low-interaction honeypots were used to study ICS attack patterns, revealing tactics such as DoS, remote code execution, buffer overflows, and payload injection, while noting the challenges of IoT honeypot deployment. In [10], tools including T-Pot, Wireshark, Honware, IoTcandyJar, and wormholes improved IoT honeypot functionality, with a call for more research using machine learning and manual analysis. In [11], a distributed honeynet emulating smart meters and AMI Headend captured attacks like False Data Injection, ARP spoofing, and Mirai, validating realistic emulation as effective for CTI.

As noted in [11], enhancing cyber-attack templates in honeypots required proper tools to track attack trends. In [9], mismatches between honeypots and PLCs were identified but not addressed. In [4], default T-Pot templates were used, making them recognizable to advanced attackers. Study [10] called for IoT-specific honeypot research, while [12] deployed a smart grid honeypot, revealing targeting patterns like initial access and reconnaissance via MITRE ATT&CK for ICS. Together, these works highlighted the value of honeypots in collecting actionable threat data.

This study presents an integrated approach that combined a high-interaction T-Pot honeypot, the Security Onion analysis platform, and the MITRE ATT&CK framework within a unified cloud-based environment tailored for IoT and SCADA systems. We generated Cyber Threat Intelligence (CTI) through data collected from a T-Pot honeypot and SCADA systems in an IoT environment. A publicly exposed honeypot was simulated, and advanced tools were integrated for event analysis and intelligence gathering. Cyberattacks were then mapped to the MITRE ATT&CK framework. This integration enhanced detection, visibility, and interpretation of adversarial activities, offering a practical improvement over previous approaches targeting critical infrastructure. Unlike prior studies that applied these tools independently, this configuration enabled end-to-end threat intelligence generation, from live attack capture and behavioral analysis to structured adversary mapping. The approach provided a practical

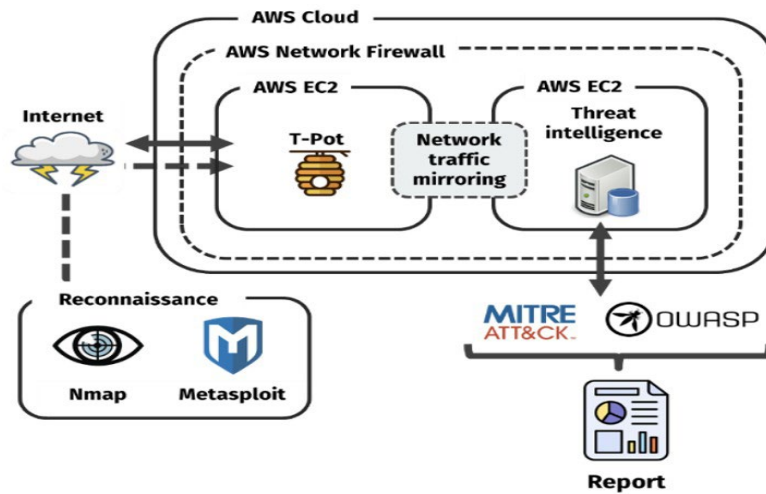


Fig. 1. Overview of the cyber threat intelligence framework using T-Pot, Security Onion and MITRE ATT&CK.

integrated model for proactive and automated CTI development in industrial and IoT contexts.

This article is organized as follows. Section II describes the methodology and tools used, implementations of the setup for cyber threat data collection, data collection timeline and analysis process using different analyzers. Section III discusses the results and insights while in Section IV, limitations are summarized. We conclude in Section V.

II. METHODOLOGY

A. Cyber Attack Data Collection – Honeypot Setup

A cloud-based honeypot environment was deployed using AWS EC2 instances, with network security configurations designed to restrict access to management ports while exposing standard IoT/SCADA services to attract attacks. As shown in Fig. 1, this setup enabled realistic threat simulation and data capture without exposing sensitive components to external actors.

As shown in Table I, security settings were configured to control inbound and outbound traffic. Access to key T-Pot services—such as Dashboards (TCP/64297) and SSH (TCP/64295)—was restricted to authorized IPs, while numerous ports tied to simulated services remained publicly open to attract attacker interactions.

TABLE I. RULES CONFIGURATION

Protocol	Port range	Source	Traffic
TCP	64295	70.68.196.3/32	Inbound
TCP	64297	50.92.164.253/32	Inbound
TCP	64297	38.25.26.68/32	Inbound
UDP	1 - 64000	0.0.0.0/0	Inbound
TCP	64297	70.68.196.3/32	Inbound
TCP	64297	199.243.207.46/32	Inbound
TCP	1 - 64000	0.0.0.0/0	Inbound
All	All	0.0.0.0/0	Outbound

B. Cyber Attack Data Analyzer – Security Onion Setup

A software platform was deployed on another EC2

AWS instance to capture network traffic from interactions between attackers and the honeypot, enabling threat intelligence analysis on raw attack data. This study utilized Security Onion, an open-source platform integrating network security monitoring and intrusion detection tools to enhance network security analysis.

Security Onion was deployed on a dedicated EC2 instance within a preconfigured VPC using an AWS Marketplace image. Two interfaces were configured—one for traffic capture and another for administration. Network and firewall settings were adjusted for secure monitoring, and a separate EC2 client within the same VPC was used to access the management interface safely.

To support real-time threat analysis, a traffic mirror session was configured between the T-Pot honeypot and the Security Onion instance. The T-Pot EC2 was the source, while Security Onion received mirrored traffic. An AWS Mirror Filter defined the scope of replication. Security Onion then enabled intrusion detection and analysis through three key interfaces detailed in Table II.

TABLE II. SECURITY ONION SECTIONS

Sections	Description
Alert	Real-time alerts from the system's intrusion detection and prevention mechanisms indicate unauthorized access attempts, malware traffic, malicious scripting, and suspicious network behavior.
Dashboard	Allows monitoring of critical security indicators, network traffic, and tracking threat trends.
Hunt	Tools for analysis of network traffic and logs, aiding root cause investigations, uncovering hidden threats, and analyzing anomalies flagged in the other Sections.

This configuration section ended with verifying network traffic capture between the T-Pot and the Security Onion by running tcpdump on the monitor interface. Table III provides an overview of the tools and techniques used in this project.

TABLE III. TOOLS AND TECHNIQUES

Tools and Techniques	Description
T-Pot Environment	A high-interaction honeypot simulating multiple protocols, using the ELK Stack for log management and Kibana for threat visualization.
Security Onion	A platform that combines tools for network security monitoring and intrusion detection to improve the analysis of network security.
MITRE ATT&CK	A systematic and comprehensive model to classify different tactics, techniques, and procedures (TTPs) utilized by adversaries during cyber-attacks.
NMAP	A program for security audits, vulnerability assessment, and network reconnaissance.
Metasploit	A framework for vulnerability evaluations, system exploitation, and security assessments.
Modbus	Protocol in SCADA environments for remote monitoring, control, and data acquisition.

C. Cyber Attack Data Collection And Analysis Process

From February 12 to March 11, 2024, cyber-attack data was gathered via T-Pot, targeting services like web, RDP, SSH, and SCADA. Analysis focused on timestamps, attacker IPs, and accessed services. To validate the setup, two steps were taken: (1) *Nmap and Metasploit reconnaissance on the SCADA honeypot to detect exposed services*, and (2) *reviewing T-Pot dashboards to confirm active threats*. Nmap scans identified a web interface for a “Pump control unit”. Further, scans targeted SCADA-related services, extracting details on PLCs and sensors via the SNMP protocol, including operational status, performance metrics, and alarms. These findings provided critical insights into the target’s configuration and potential vulnerabilities.

Metasploit was employed to detect Modbus-enabled devices to enumerate Modbus Unit IDs, revealing the SCADA network structure. These IDs identify individual nodes and offer system layout insights. Relevant CVE data was also sorted and cleaned automatically to extract key details like timestamps, attacker IPs, accessed services, and activity logs.

From March 23 to 31, 2024, T-Pot captured attack data across web, RDP, SSH, and SCADA services, focusing on timestamps, attacker IPs, accessed services, and activities. Intelligence from selected sources was integrated and mapped to MITRE ATT&CK techniques. Key incidents were analyzed to identify persistence, initial access, reconnaissance, and coordinated attack patterns.

The T-Pot honeypot, configured with SCADA/IoT modules, was deployed on an EC2 instance to emulate real-world attacks. Nmap and Metasploit validated attack visibility, while T-Pot dashboards confirmed adversarial activity. Integrated with Security Onion using custom mirroring filters, the system enabled real-time monitoring. Captured data was mapped to the MITRE ATT&CK framework, and dynamic dashboards were built to analyze key security events.

The initial T-Pot honeypot environment was deployed

on AWS and ran continuously for approximately one month to collect baseline cyber-attack data. However, due to required environment reconfiguration, only eight days of integrated operation with the Security Onion were feasible for the final dataset. Establishing and validating the environment, including EC2 instance provisioning, traffic mirroring, and firewall configuration, took roughly two weeks of preparation and testing. The total AWS cost for this deployment, including compute, storage, and data transfer, was approximately USD 300. Extending the collection period in future work would enable a more robust identification of long-term threat patterns, campaign evolution, and periodic attacker behaviors across different time zones.

III. RESULTS & DISCUSSIONS

This study generated CTI by collecting data through a T-Pot honeypot and SCADA systems for IoT environments. Advanced insights were delivered through dynamic dashboards using Security Onion and the MITRE ATT&CK framework. The study simulated an internet-facing honeypot, integrated analysis tools, and developed a cloud-based dashboard for interactive visualization. Processed intelligence was mapped to MITRE ATT&CK to analyze and interpret cyberattacks effectively.

Intelligence was gathered by integrating and interpreting selected sources to support strategic security decisions. The initial analysis used 30 days of T-Pot data, followed by 8 days of the combined T-Pot and Security Onion setup. Attacker actions were mapped to relevant MITRE ATT&CK techniques, and key incidents were analyzed to identify traits such as persistence, initial access methods, reconnaissance, and coordinated campaigns.

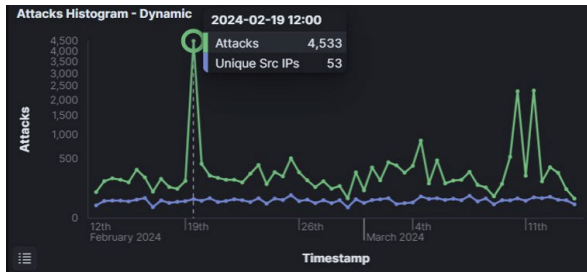
A. T-Pot Cyber Attack Data Analysis

Over 30 days, the honeypot collected valuable data from the attacker’s interactions. The ConPot recorded 16,143 attempted attacks originating from an array of 1,446 unique sources of IP addresses. An immediate remark on this number of attacks is that emulated SCADA services running on ConPot became high-profile targets. It drew the attention of a considerable number of actors, looking to exploit vulnerabilities within the realm of devices that played a role in controlling key industrial processes, such as maintaining and managing.

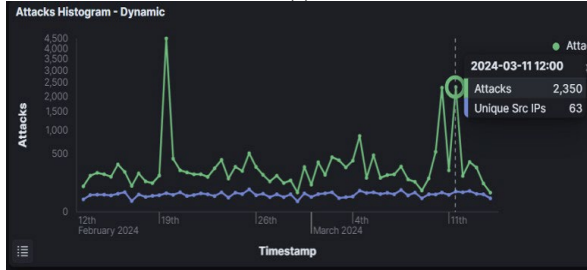
1) *Attack Histogram*: The attack histogram analysis revealed activity variations over a month of operations.

On 2/19/2024, a spike of 4,533 attacks from 53 unique source IP addresses was observed, as shown in Fig. 2 a). Subsequent days exhibit fluctuating attacks until 03/10/2024 (2326 attacks from 57 unique IP addresses) and 03/11/2024 (2350 attacks from 63 unique IP addresses), as shown in Fig.2 b). These attacks suggest actions targeting specific vulnerabilities or pivots in malicious activity.

2) *Source Ip Address Reputation*: The source IP address reputation from attackers provided insights into the nature of the observed attacks. Of the attacks, 3867



(a)



(b)

Fig. 2. Attack histogram: a) Peak 1, and b) Peak 2 and 3.

originated from "Known Attacker" IP addresses, indicating a persistent threat from established malicious sources. Additionally, 1411 instances were identified as "Mass Scanner", suggesting an automated scanning engine behind such activities, likely looking for vulnerabilities throughout a range of targets, as shown in Table IV.

TABLE IV. DISTRIBUTION OF ATTACKER SOURCE IP REPUTATION

Attacker Source IP Reputation	Count	Percentage
Known attacker	3867	73%
mass scanner	1411	27%
anonymizer	11	0.2%
bot, crawler	11	0.2%
tor exit node	2	0.0%

Notably, 11 attacks originated from "Anonymizer" IPs, indicating efforts to conceal the source and complicate threat identification. Another 11 attacks were linked to "bot, crawler" activity, suggesting automated agents. Additionally, two attacks came from "Tor Exit Nodes", highlighting the use of the Tor network to obscure malicious traffic and further hinder tracking.

3) *Protocols Detected*: Honeypot protocol analysis revealed diverse malicious activity. HTTP led with 14,045 events, indicating web-based attacks such as reconnaissance and brute force, followed by FTP with 3,943 events. SCADA protocols were actively targeted—S7comm (1,196), enip (514), IEC104 (233), along with Modbus (1,103), highlighting continued focus on industrial control systems.

Table V shows IP interactions with the honeypot, highlighting the top 10 ASNs and the most active attacker IPs. IP 109.123.234.41, linked to AS141995 (Contabo Asia), accounted for 4,292 events. While the dataset includes diverse attack sources, some ASNs may reflect

TABLE V. TOP ASN SOURCES BY COUNT AND IP ADDRESS

ASN	ASN Count	Source IP	IP Count
Contabo Asia	4295	109.123.234.41	4,292
Digital Ocean	3217	70.68.196.3	1,519
Shaw	1521	159.223.86.219	562
Google Cloud	1402	185.254.196.173	405
OVH SAS	642	165.22.101.81	279
Ucloud	538	152.32.226.8	203
Virtual Systems	521	165.154.206.241	203
Amazon	367	135.125.244.48	188
Hurricane	341	135.125.217.54	164
SCLoud Pte Ltd	223	135.125.246.110	136

cloud providers or ISPs, requiring deeper analysis to distinguish legitimate services from malicious activity for accurate threat attribution.

B. T-Pot And Security Onion Cyber Attack Data Analysis Via Mitre Att&Ck

The integrated solution, consisting of T-Pot SCADA and Security Onion, was operational and available within the AWS environment for 8 days, from March 23 to 31, 2024. Through-out this period, it effectively captured multiple cyberattacks.

The final phase involved developing a series of dashboards to facilitate exploring security events. Integrating and interpreting intelligence from the selected sources helped us gain comprehensive insights and map these findings to relevant MITRE ATT&CK techniques. MITRE ATT&CK framework provides businesses valuable insights into cybersecurity threats, weaknesses, and trends to recognize, assess, and successfully counteract cyber threats. Tactics refer to the various activities cybercriminals employ to execute an attack. Analyzing TTPs enables security teams to identify and counteract threats by comprehending the methods used by attackers.

Some important threat intelligence key insights were:

- Total Of 39,412 events were generated during the observation period. The top 5 attack alerts were identified across high, medium, and low severities. The top attack alerts by volume are predominantly of medium severity, with nearly 84% of all alerts. In addition, 0.5% of these events were related to high-severity alerts. Nonetheless, due to their critical nature, these high-severity alerts necessitated further analysis; details of top alerts are shown in Table VI.

TABLE VI. TOP 5 ATTACK ALERTS BY PERCENTAGE

Top 5 Attack Alerts	Percentage
ET DROP Dshield Block Listed Source	38%
ET CINS Active Threat Intelligence Poor Reputation IP	26%
ET INFO External IP Address Lookup Domain (ipify.org)	13%
Other	23%

- Table VII and Table VIII show a broad spectrum of port usage and underscore the need for comprehensive monitoring and security measures across all ports. The analysis revealed that 79% of network traffic targeted non-standard ports, likely due to the honeypot server running various services.

TABLE VII. DISTRIBUTION OF DESTINATION PORTS BY PERCENTAGE

Destination Ports	Percentage
HTTPS (443)	13%
SSH (22)	3%
HTTP (80)	2%
Potential Misuse (0)	2%
MySQL (3306)	1%
Other ports	79%

- Among the remaining traffic, 13% was directed to port

TABLE VIII. TOP SOURCE AND DESTINATION IPS WITH NUMBER OF ATTACKS

Top Source IPs	# Attacks	Top Destination IPs	# Attacks
172.31.10.207	11971	172.31.10.207	86152
185.161.248.148	1107	104.26.13.205	3878
94.232.45.94	1098	172.67.74.152	3878
149.28.141.48	731	104.26.12.205	3771
52.207.228.53	338	34.117.186.192	40

443 (HTTPS), commonly used for secure web communication. Port 22 (SSH), associated with secure shell access, accounted for 3%, while port 80 (HTTP), used for unencrypted web traffic, received 2%. Another 2% targeted Port 0, typically invoked when a client requests a dynamically assigned port.

During this phase, key incidents were analyzed to identify traits such as persistence, initial access, reconnaissance, and coordinated campaigns. The study examined indicators, threat definitions, sources, motivations, impacts, and MITRE ATT&CK mappings for two main threat groups: *a) high-severity* and *b) high-volume intrusion activity*, as shown in Table IX that outlines Suricata's alert classifications.

TABLE IX. SURICATA'S ALERTS AND DEFINITIONS

Alerts	Description
High severity	Events that pose significant risks to network or system security, such as exploit attempts or malware activity.
Medium severity	These events may moderately impact network or system security, often involving anomalies or policy violations.
Low Severity	These events involve protocol anomalies or misconfigurations that pose no immediate threat but still warrant investigation.
High Volume	These alerts signal security events, unusual activity, or threats that require immediate action, such as network issues or frequent alert spikes.

To ensure a consistent mapping process, a structured review process was applied to all detected alert categories during the MITRE ATT&CK correlation. Each alert signature was cross-checked with the official MITRE ATT&CK knowledge base to identify the most suitable tactic and technique. The classification was based on the alert description and observed attack behavior captured through the honeypot analysis. When multiple techniques were possible, the mapping selected the one that best represented the main or recurring adversarial behavior in the dataset.

a) Threat Group: High-Severity Intrusion Activity:

High-severity alerts accounted for 425 alerts, as depicted in Table X. From the analyzed high-severity alerts, 299 of the alerts belonged to three categories, as shown in Table XI. In addition, as seen in Table XII, a key insight involved the geographic origin of source and destination IPs. This suggests implementing country-based filtering or regional restrictions. For example, limiting access to North America for services offered only in Canada or the U.S. could significantly reduce attack frequency.

TABLE X. DISTRIBUTION OF TOP ALERT CATEGORIES

Top alert categories	Count	Percentage
Possible MS CMD Shell opened on local system	178	42%
External Oracle T3 requests inbound	80	19%
Possible Linux Mirai login attempt	41	9%
Other categories	126	30%

Each intrusion follows a purpose and a plan [13]. Based on the observed tactics described, the attacker's objective could be disrupting operations, exfiltrating data, or integrating the T-Pot SCADA into a botnet. As described in [12], a successful intrusion could lead to loss of control over the SCADA system, data breaches, or disruption of critical infrastructure, causing business interruptions and possibly compromising vital assets.

- *MITRE ATT&CK Mapping:* The top three high-severity alerts were manually mapped to MITRE ATT&CK using keyword matching, as detailed in Table XIII. Mapping began by analyzing alert categories and extracting key terms like "Command Shell," "External Request," and "Login Attempt" to identify potential tactics or techniques. Since alerts may relate to multiple

TABLE XI. HIGH VOLUME ALERTS SUMMARY

Category	Summary
Misc Attack - DROP DShield Block Listed Source Rate: 38%	Indicates traffic from IP addresses listed by the DShield Block List, often associated with botnets, malware, and scanning activity. "ET" (Emerging Threats) signatures flagged these sources based on collective intelligence. Behavioral patterns suggest alignment with botnet-driven campaigns such as Mirai, which exploit IoT and ICS devices to amplify DDoS attacks and propagate across vulnerable networks.
Misc Attack - CINS Active Threat Intel. Poor Reputation IP Rate: 26%	Detects activity from IP addresses identified by the CINS Active Threat Intelligence feed as having a poor reputation. These IPs are linked to known malicious behavior including spam, malware delivery, and exploit attempts. Events in this category suggest broad scanning operations and automated exploitation tactics used to identify and compromise vulnerable systems.
Misc Activity - External IP Address Lookup Domain in TLS SNI Rate: 13%	TLS connections containing domains like ipify.org in the SNI field signal external IP resolution, a reconnaissance method for environment mapping. This behavior aligns with APT tactics involving network discovery and host profiling prior to targeted exploitation.

TABLE XII. SOURCE AND DESTINATION COUNTRIES BY COUNT

Source country	Source Count	Destination country	Dest. count
United States	81	United States	100
India	37	India	28
China	28	The Netherlands	19
United Kingdom	25	Singapore	14
The Netherlands	20	Australia	14
Others	234	Others	250

TABLE XIII. MITRE ATT&CK MAPPING OF HIGH SEVERITY TOP ALERTS

TTPs	MITRE ATT&CK mapping	Description in this Con-text
Command Shell	TA0002 Execution > T1059 Command and Scripting Interpreter >> T1059.003 Windows Command Shell	Establishing command shell could be used to move laterally within the network.
T3 Exploit	TA0001 Initial access > T1133 External remote services	Targeting vulnerabilities in Oracle T3 protocols could be an initial access attempt.
Mirai Login Attempt	TA0011 Command and control > T107 Application layer protocol >>T1071.001 Web protocols	Attempted logins using Mirai malware could be another method for lateral movement or privilege escalation.

techniques, cross-referencing with MITRE ATT&CK descriptions ensured accurate classification.

- *Mitigation:* To mitigate these attacks, immediate action was necessary due to the high event volume within 8 days. Isolating the T-Pot SCADA system helped prevent lateral movement and further compromise the environment. Future investigations should include forensic analysis to determine root causes and attacker intent. Additionally, T3 protocol vulnerabilities must be patched, and security reinforced through network segmentation, access controls, and IDS/IPS deployment.

b) Threat Group: High Volume Of Potential Intrusion: High-volume alerts accounted for 77% of all alerts, totaling 39,412. They were grouped into three main categories, as depicted in Table XIV.

TABLE XIV. TOP ALERTS BY COUNT

Top Alerts	Count
Dshield Block Listed Source	14843
Active Threat Intl. Poor Reputation IP	10293
External IP Lookup Domain in TLS SNI	5060
Other	9216

Security Onion identified and categorized these alerts from the T-pot SCADA within the research timeframe. The 30196 alerts belonged to the three categories described in Table XV.

TABLE XV. HIGH-SEVERITY ALERTS SUMMARIZED

Category	Summary
ET ATTACK RESPONSE Possible MS CMD Shell opened on the local system. Rate: 42%	This category indicates potential attempts to establish a command shell on the local system. Command shells provide attackers with a powerful interface to execute commands and manipulate the system.
ET POLICY External Oracle T3 Requests Inbound. Rate: 19%	This reflects unauthorized attempts to access Oracle T3 services in the T-Pot SCADA setup, likely through middleware exploits for remote access or data exfiltration—common in coordinated attacks on critical infrastructure.
ET MALWARE Possible Linux Mirai Login Attempt. Rate: 9%	This indicates brute-force login attempts by the Mirai botnet, which exploits weak authentication in Linux-based IoT devices to recruit them for DDoS attacks. Evolving variants continue to drive large-scale campaigns against IoT and ICS networks globally.

Intrusion attempts on the T-Pot SCADA honeypot exposed layered attack strategies using malware, botnets, exploits, spam, and IP resolution. High-volume alerts (77%) indicated deliberate threat activity [13]. Mapping to MITRE ATT&CK and Table XV revealed goals such as espionage, sabotage, data theft, reconnaissance, and botnet growth, typical of SCADA-targeted campaigns. These threats risk operational disruption, data loss, and infrastructure compromise [12].

- *MITRE ATT&CK Mapping:* As shown in Table XVI, high-volume intrusion alerts were mapped to the MITRE ATT&CK framework through manual analysis. Since one alert can align with multiple tactics or techniques,

TABLE XVI. MITRE ATT&CK MAPPING

TTPs	MITRE ATT&CK mapping	Description
Exploit Attempts	TA0001 Initial Access > T1189 Drive to Compromise TA0002 Execution > T1059 Command and Scripting Interpreter TA0011 Command & Control > T1132 Data Encoding	Infection of devices when visiting compromised or malicious websites. Use of scripting languages or command interpreters to execute malicious commands. Encode of communications to obfuscate command and control (C2) traffic.
Poor Reputation IP	TA0040 Impact > T1485 Data Destruction TA0007 Discovery > T1082 System Information Discovery TA0005 Defense Evasion > T1027 Obfuscated Files or Information	Execution of command-and-control infrastructure, initial access vectors, and data exfiltration methods.
IP Address Lookup (SSL/TLS)	TA0011 Command & Control > T1071 Standard Application Layer Protocol TA0007 Discovery > T1572 Protocol Tunneling TA0006 Credential Access > T1557 Man-in-the-Middle	C2 communications with HTTPS to blend in with legitimate network traffic. Hide malicious traffic within encrypted channels. Steal sensitive information such as usernames, passwords, or authentication tokens.

the mapping reflects a many-to-many relationship, offering a structured approach to interpreting the identified threats.

- **Mitigation:** IP addresses with poor reputations should be blocked immediately. To attract attackers, the T-Pot honeypot can be enhanced with new vulnerabilities and deception tactics, like weak authentication. Behavioral analysis helps uncover attacker methods and IoT targets. Sharing anonymized data with other operators improves threat visibility. Measures such as network segmentation, access controls, and IDS/IPS should be implemented to strengthen the T-Pot SCADA system.

IV. CHALLENGES, LIMITATIONS & FUTURE WORK

The initial T-Pot honeypot environment was deployed on AWS and ran continuously for approximately one month to collect baseline cyber-attack data. However, due to required environment reconfiguration, only eight days of integrated operation with the Security Onion were feasible for the final dataset. Establishing and validating the environment, including EC2 instance provisioning, traffic mirroring, and firewall configuration, took roughly two weeks of preparation and testing. Extending the collection period in future work would enable a more robust identification of long-term threat patterns, campaign evolution, and periodic attacker behaviors across different time zones.

While this study shows that honeypots are effective for cyber threat intelligence, limitations exist. Advanced attackers may detect the environment due to limited realism, such as static setups or a lack of user activity, which can reduce the chance of capturing complex tactics. Future deployments should adopt adaptive deception and dynamic traffic simulation. Additionally, deploying the honeypot on AWS may lead to detection biases, as attackers can recognize cloud signatures and alter their behavior. This likely explains the high volume of reconnaissance alerts and fewer exploitation attempts. The 8-day window limits visibility into long-term threats, which a longer deployment could better capture.

In future, we aim to improve the MITRE ATT&CK mapping process by using automated correlation and contextual analysis beyond of keyword matching. Machine learning and semantic analysis techniques will be explored to identify attacker tactics and techniques more accurately from raw intrusion data. At the same time, the project will consider developing a unified CTI framework that enables adaptive data enrichment, information sharing between platforms, and automated response workflows.

V. CONCLUSIONS

In this research, we used a honeypot to collect and analyze cyberattacks, patterns, and trends in an IoT environment. The individual components employed in this study, T-Pot, Security Onion, and the MITRE ATT&CK framework, are well-established in cybersecurity research, their coordinated orchestration within a unified workflow constitutes the principal contribution of this work. The

attack data enabled threat detection, analysis, and intelligence gathering while exposing vulnerabilities in industrial control systems, which attackers exploited immediately. The integration architecture facilitated data correlation, adversarial mapping, and contextual enrichment of cyber threat intelligence, demonstrated the significantly enhanced analytical depth and operational effectiveness in CTI processes. The findings highlight how organizations can strengthen defenses by using honeypots to study attacker strategies and adopt a proactive cybersecurity approach. Future enhancements will provide an automated system to capture and analyze cyber threat intelligence from honeypot environments to more accurate, actionable, and valuable for real-world defense mechanisms.

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